

Foundation (Jalapeno)

- Q1.** If two lines have the same slope, what can be said about them?
(A) They are perpendicular
(B) They are parallel
(C) They are intersecting
(D) They are skew
(E) They are coincident
- Q2.** A basketball player scores $2x + 5$ points in the first half and $2x - 3$ points in the second half. If the lines representing these scores are parallel, what is the value of x ?
(A) -4
(B) 0
(C) 2
(D) 4
(E) 6
- Q3.** The equations of two parallel lines are $y = 3x + 2$ and $y = 3x +$
(A) -2
(B) 0
(C) 2
(D) 4
(E) 6
- Q4.** A cyclist travels $5x - 2$ miles in the morning and $5x + 1$ miles in the afternoon. If the lines representing the distances traveled are parallel, what is the slope of the lines?
(A) 1
(B) 2
(C) 3
(D) 4
- (E) 5
- Q5.** If the equation of a line is $y = 2x - 1$, what is the equation of a parallel line that passes through the point $(2, 3)$?
(A) $y = 2x + 1$
(B) $y = 2x - 3$
(C) $y = x + 1$
(D) $y = x - 1$
(E) $y = 3x - 2$
- Q6.** The graph of $y =$
(A) $2x + 1$
(B) $2x - 1$
(C) $x + 2$
(D) $x - 2$
(E) $3x - 1$
- Q7.** A tennis player wins $4x + 2$ games in the first set and $4x - 1$ games in the second set. If the lines representing the number of games won are parallel, what is the value of x ?
(A) -1
(B) 0
(C) 1
(D) 2
(E) 3
- Q8.** If the equation of a line is $y =$
(A) $3x + 2$
(B) $3x - 2$
(C) $2x + 1$
(D) $2x - 1$
(E) $x + 3$

Open Ended Questions (FRQ)

- Q9.** A golfer hits a ball that travels $2x + 1$ yards on the first shot and $2x - 2$ yards on the second shot. Find the equation of the line that represents the total distance traveled by the ball after x shots, if the lines representing the distances are parallel.
- Q10.** A runner completes a 5-mile course in x hours. If the runner's pace is constant, find the equation of the line that represents the distance traveled after x hours, and explain why the lines representing the distance traveled at this pace are parallel.

Chef's Tips

- Tip 1: Encourage students to use the concept of slope to identify parallel lines.
- Tip 2: Remind students to use the equation of a line in slope-intercept form to find the equation of a parallel line.
- Tip 3: Emphasize the importance of analyzing relationships between parallel lines in real-world contexts, such as sports and fitness.